p61 selected-class isogeny census

Certified launch evidence and completion boundary

Field	`p = 2^61 - 1 = 2305843009213693951`
Selected root	`p61-i0025`, trace `-704859453`
Class order	`358094304`, factors `[179047152, 2]`
Run	`p61-class-pilot-v2-20261014T0902Z`
Evidence date	2026-10-14

CLASS ORDER CERTIFIED PREFIX AUDIT PASSED FULL CENSUS INCOMPLETE

Scope

One selected ordinary isogeny class over the field. This report does not claim enumeration of every curve or every isogeny class over the field.

Source commits: producer `c9bee1b2`; auditor `1e9f40817`; parallel replay candidate `d06472954`; certification runner `d2f5f49a`.

1. Decision

PARI `bnfcertify(bnf)` with flag 0 returned `1` after 10,365.337 seconds. The dedicated receipt unconditionally certifies the selected fundamental discriminant `-8726545188371316595` with class number `358094304` and cyclic factors `[179047152, 2]`.

The complete bounded pipeline has been exercised on an immutable 300,672-record prefix: generation, independent mathematical replay, checkpoint and hash-chain validation, dual-evidence ledger sealing, external sort, and uniqueness audit. The certified audit found 300,672 unique j-invariants and zero duplicate records.

Current boundary

Coverage is 300,672 of 358,094,304 vertices, approximately 0.0839645%. The receipt status is `partial_census_pass`; `complete_class_coverage` is false. The degree 5/13/37/59 adjacency pass is separate and pending.

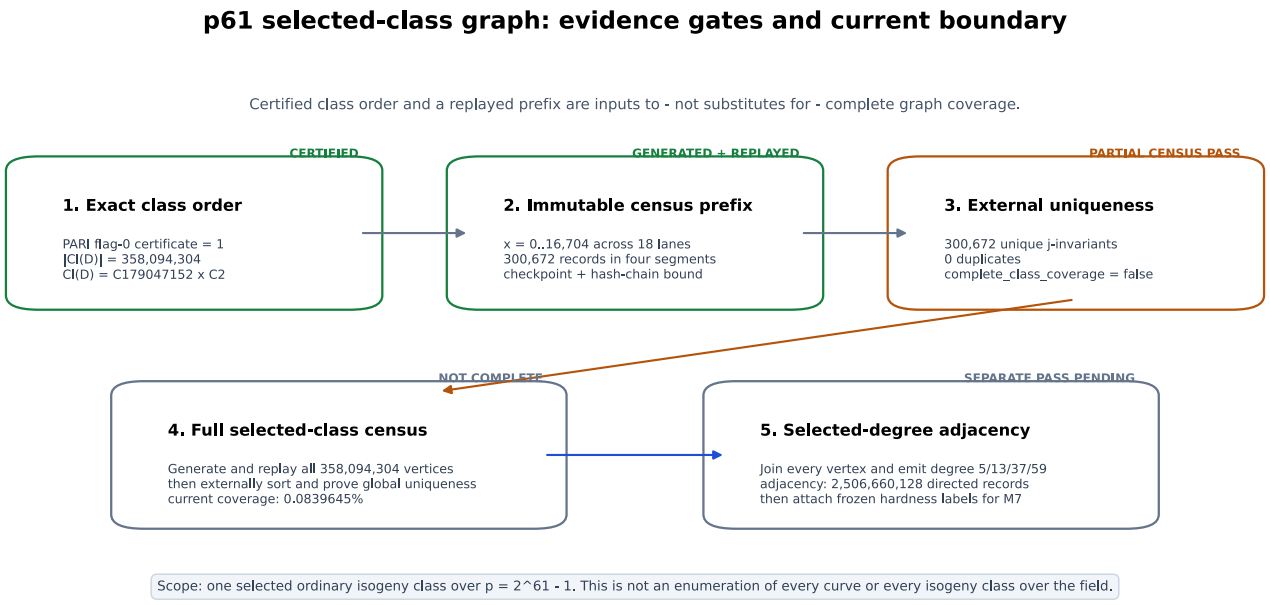


Figure 1: Certified inputs, completed prefix gates, and the remaining full-census and adjacency gates.

2. Requested outcome and evidence state

Component	State	Evidence boundary
Exact selected-class order	CERTIFIED	Flag-0 certificate returned `1`; receipt SHA-256 `260382fb...f9643c9`.
Bounded-memory producer	IMPLEMENTED	Committed Rust producer with checkpoints, chained records, compressed segments, and failure retention.
Independent replay	PASS	All four segments mathematically replayed with distinct order-audit settings.
External uniqueness	PARTIAL PASS	300,672 records sorted; 300,672 unique invariants; zero duplicates; incomplete class range.
Full vertex census	INCOMPLETE	300,672 of 358,094,304 vertices.
Selected-degree adjacency	PENDING	2,506,660,128 expected directed degree 5/13/37/59 records after the full join.
All field-wide curves/classes	NOT ENUMERATED	A different and much larger objective than completing this selected ordinary class.
M7 large-field use	NOT ADMISSIBLE YET	Requires complete graph evidence, frozen labels and masks, and leakage-safe successor controls.

2.1. Mathematical identity

Quantity	Value
Prime	`2305843009213693951`
Trace	`-704859453`
Fundamental discriminant	`-8726545188371316595`
Conductor	`f_pi = 1`
Class number	`358094304`
Cyclic factors	`[179047152, 2]`
Certification	`bnfinit(polynomial, 1)` then `bnfcertify(bnf)`
PARI flag/result	`0 / 1`
PARI version	`2.17.2`
Wall time	`10365.721123667 s`

3. Certified prefix generation and replay

The traversal has 18 lanes over 19,894,128 columns. The current prefix contains two 32-column pilot segments, one 256-column expansion, and one 16,384-column certified expansion.

Segment	Vertices	Generate	c9 replay	d064 replay
`x = 0..32`	576	13.415 s	14.778 s	23.204 s
`x = 32..64`	576	13.421 s	14.672 s	22.693 s
`x = 64..320`	4,608	21.266 s	46.939 s	33.460 s
`x = 320..16,704`	294,912	858.472 s	not run	735.772 s
Total	**300,672**	**906.574 s**	**76.389 s†**	**815.129 s**

The c9 total excludes the largest segment. The committed `d06472954` parallel replay candidate passed all four immutable segments and reproduced their record-chain, checkpoint, and segment digests. It recorded 23.204 s and 22.693 s on the short segments: 1.570x and 1.547x the earlier replay times. On the 4,608-record expansion, the first comparison was 33.460 s for d064 versus 46.939 s for c9, a 1.403x c9/d064 ratio. A second c9 control took 215.940 s during several release compiles. Every identity agrees, but this variance precludes a general speedup claim; controlled isolated ABBA runs remain required.

On the 294,912-record expansion, generation took 858.472 s and d064 replay took 735.772 s (343.531 and 400.820 records/s). No matched c9 replay was run there, so this establishes sustained operation and correctness, not a speedup.

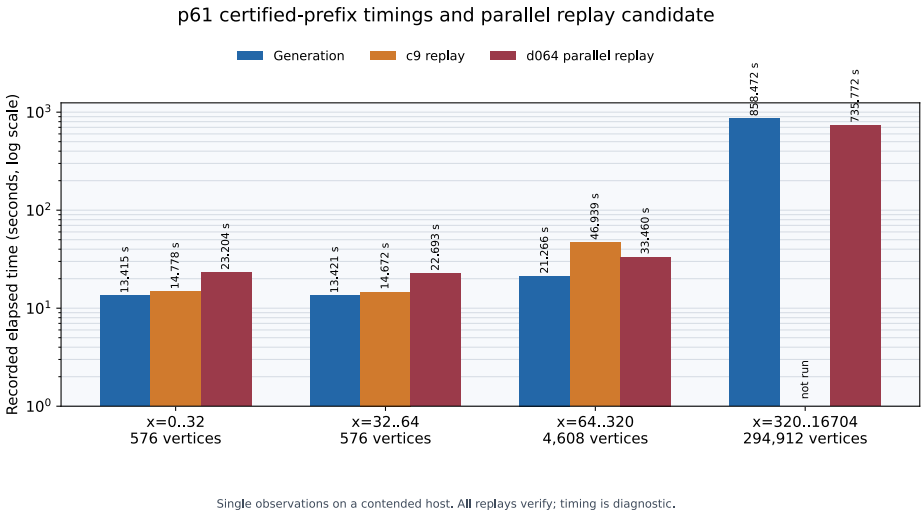


Figure 2: Startup-inclusive prefix measurements. One observation per bar on a shared, unisolated host.

Extrapolation from the 16,384-column expansion gives about 12.1 days for generation and 10.3 days for d064 replay, or 22.4 days for those phases alone. These are diagnostic projections, not schedule commitments; audit, interruptions, and adjacency are additional.

4. Certified ledger and durability

The dedicated certificate binds the exact provisional preflight receipt by SHA-256. Existing segment receipts retain their original preflight binding; the v2 ledger adds the certificate as a second immutable dependency instead of rewriting historical evidence.

Audit quantity	Result
Ledger range	`x = 0..16,704`
Records	300,672
Unique j-invariants	300,672
Duplicate records/groups	`0 / 0`
Complete class coverage	`false`
Status	`partial_census_pass`
Sorted-index bytes	4,810,768
Sorted-index SHA-256	`055fb9e007b0b7a261f4850b78450d1d8f306ec56e13c316cebf20514d1793ad`

4.1. Durable storage semantics

The active UGREEN SMB export is mounted at `/Volumes/data`; the requested alias `/Volumes/UgreenNAS` is not mounted. Artifacts live under `/Volumes/data/ec-hardness-atlas-20261012/`.

The SMB mount reports `sync_all_unsupported_by_filesystem` for derived ledger, chunk, index, and receipt files. That limitation is recorded rather than relabeled. A host `sync` barrier completed, then all terminal digests were replayed. The certification runner's receipt reports `fsync_completed`.

4.2. Preserved failures and controls

- The first class-number attempt failed during `bnfinit` with PARI's low-precision `small_norm` error. Its event log and failure receipt remain.
- The successful runner retried at 256 digits and retains the initial error.
- The flag-1 quotient-only control was interrupted after it remained non-decisive; its control receipt remains content-addressed.
- Interrupted segment and SMB per-file-sync attempts remain under `p61-class-streaming-v1/incomplete/` and are not admitted.
- The first `d06472954` release build failed during archive creation because the system data volume had only 187 MiB free. It remains classified as an infrastructure failure in `build-attempts.json`; the corrected build uses a scoped target on SSD990-2. After that build passed, only the generated `/private/tmp` target was deleted, restoring local headroom to 5.5 GiB.
- The auditor rejects an unconditional label without the dedicated flag-0 certificate. The false declaration exited nonzero and produced no accepted ledger.
- Two default-parallel whole-library release-test attempts remained non-green in the pre-existing `jv_cover` module (2 and 3 failures respectively); every exposed failure passes in exact isolation and the p61 auditor gate passes 6/6. A full low-concurrency run with `--test-threads=1` passed 4,079/4,079, with 114 ignored, in 2,765.68 seconds. The varying default-parallel failures and explicit internal timeout remain in `test-results.json`; those attempts are not relabeled as passes. The library is green under the recorded serial mode, while default-parallel execution remains non-green on this shared host.

5. Storage and scale boundary

At the post-expansion observation, the UGREEN export reported 7.3 TiB total, 198 GiB used, and 7.1 TiB available.

Quantity	Estimate	Interpretation
All selected-degree directed records	2,506,660,128	Seven directed degree 5/13/37/59 neighbors per class vertex.
Raw JSON projection	about 5.14 TB	Dominated by the later adjacency representation.
Projection plus 25% headroom	about 6.43 TB	Fits only narrowly on the current active volume.
UGREEN available	7.1 TiB	Insufficient for redundancy, multiple large fields, and broad intermediate products.
Requested petabyte tier	not provisioned	Requires separate object or cold storage plus lifecycle and replication policy.

Capacity decision

The UGREEN volume can serve as the active staging tier for one selected-class campaign. It is not a petabyte store. A larger program needs packed, indexed graph bundles and separately provisioned durable storage before data generation expands to many large fields.

Even a provisioned petabyte would not support a literal field-wide census. Storing only one packed 64-bit j-invariant for each of the `2^61 - 1` possible j-values already requires `18,446,744,073,709,551,608` bytes (about 18.45 EB, or 16 EiB), before twists, models, edges, labels, indexes, replicas, or work space. A petabyte program can cover a very broad selected-class panel, not every curve over this field.

The current compressed vertex-prefix sample is much smaller than the raw adjacency projection, but its byte rate must not be transferred to the later edge corpus because the schemas and record multiplicities differ.

6. Relationship to M7

The companion M7 implementation uses class-held-out and whole-field-held-out splits and explicit masks for exhausted, skipped, capped, unverified, and incomplete-walk observations. The frozen external-field run improves the ceiling/size baseline, but its preregistered specificity gates fail.

Probe	Result	Meaning
Primary geometry-residual MLP	12.8088%	Baseline test SSE removed; interval 11.0459% to 13.9280%.
Matched geometry permutation	19.8997%	A destructive correspondence control improves more than the primary model.
Typed-edge rewiring	12.5639%	Topology destruction retains nearly the primary effect.
Node-feature shuffle	6.1334%	Node correspondence destruction retains prediction.
Excluded `log2_q` ridge reconstruction	`R2 = 0.860425`	Numeric size leakage remains recoverable.

The defensible result is stable predictive correlation for the measured labels, not isolated geometry-specific incremental information. The p61 graph is an infrastructure and scale-transfer input; it does not make the M7 specificity and leakage gates pass.

7. Completion contract

Full selected-class completion requires all seven stages:

1. Generate all 19,894,128 traversal columns across 18 lanes.
2. Independently replay every admitted segment with distinct audit settings.
3. Seal a ledger covering exactly 358,094,304 records.
4. Externally sort every invariant and prove global uniqueness and exact count.
5. Build and verify the complete degree 5/13/37/59 adjacency join.
6. Attach frozen hardness contexts, censor masks, and provenance.
7. Train and evaluate a leakage-safe M7 successor on held-out classes and fields with frozen destructive controls.

> **Non-substitution rule**
>
> A certified class order does not certify graph coverage. Prefix uniqueness does not certify global uniqueness. A full vertex census does not include adjacency until the join is run. A large graph does not by itself establish a geometry-specific hardness effect.

Enumerating every curve or every isogeny class over the field is a different, far larger objective and is not implied by completing this class.

The fail-closed continuation runner has a tested plan beginning at `x = 16,704`; future chunks are not represented as generated, replayed, or admitted.

8. Reproduction map

- Producer protocol: `docs/p61-isogeny-class-streaming.md`
- Auditor protocol: `docs/p61-isogeny-audit.md`
- Producer CLI: `src/bin/p61_isogeny_class.rs`
- Traversal implementation: `src/cryptanalysis/isogeny_walk/p61_class.rs`
- Auditor CLI: `src/bin/p61_isogeny_audit.rs`
- Fail-closed continuation runner: `research/p61_isogeny_class_20261014/run_remaining_census.py`
- Raw timing record: `timings.json`
- Timing renderer: `plot_timings.py`
- Evidence diagram renderer: `plot_evidence_pipeline.py`
- Release build attempts: `build-attempts.json`
- Release test results: `test-results.json`
- Parallel-verifier release manifest: `parallel-verifier-release-manifest.json`
- Terminal certificate manifest: `certification-terminal-manifest.json`
- Expanded prefix manifest: `certified-prefix-x000000000000-x000000016704-manifest.json`

9. Digest appendix

Artifact	SHA-256
Provisional preflight receipt	`44e202b8eb9cdad3ef09fb50e416cc8bdcf0942a6f140e24db587a3c4d44493f`
Certification receipt	`260382fbce4805ce84ce00fc77baf27c95243220ae88ca7725e31e4f4f9643c9`
Terminal manifest	`372a304bd30be4aa638264edaf84711d374118637343abd758f0cf00b951b0b1`
Generation producer binary	`47fa827fa9877a91f8b6d496f688f1df6fb18fd25a44b303392ba555be79043f`
Pilot segment `0..32`	`2a29068d66cee5914a5960bff4a13d55dd97ec80d8b9604e4f82c1a6c9c7ddf7`
Pilot segment `32..64`	`70ef588e21200e3c059695ffeae98b29faf0be702b8a01b88b0e12f888bb9fc2`
Pilot completion manifest	`84e597f9f07e674c732e39cabd88d47bf8e902cbeb083492a1e7757bcbf06b8c`
Auditor binary	`bd99626fb402f0e71587b350912f57df021ee5449791ea91bee0a8d2d3917c05`
Parallel replay binary	`81c1a5b8897bcc39fd20c6427e17eb73d9160d838bb001f82810de4272579e82`
Parallel release manifest	`a875ea679bcf04af623b28790baf8f61f0df1f64da7035506bd51da750562268`
Parallel receipt `0..32`	`b24697c9e8fbb53e594864406c3f4c98a6aeb5cd2d349b9c3b3951e0e06f79a3`
Parallel receipt `32..64`	`696d05717de484db6fe2dd92abbf6cf5fd2563e565555d7f560351cb7f5a0a8d`
Certified segment `64..320`	`b0e0180043a3b5aff0a22fa9c0a6151cfbaa0a331b62253f9aa49a97161d8852`
Generation receipt `64..320`	`d188864b5b69a3b51e3a66856a1db421941ac757a33304dbcfedbd6f1516116f`
d064 replay `64..320`	`2b11175e2cd775cac030fa5e38ef11f67b536a42a69a56bd01c646500f0da8a6`
Certified segment `320..16,704`	`514e8b0d6236ab883eb11f4cf24d8aeb66ce2a3ef6efb41652fcdd9a4fa08961`
Generation receipt `320..16,704`	`99f09bdad81ff06641063e1cdbc4d3b106622c97d9685b30c218bf873889ea0f`
d064 replay `320..16,704`	`93a36eeccdf6b821d4ffd8bebd552975ea2e0daa826672b2eb0dc88f95751dd8`
Expanded certified ledger	`822f6b0908432279351791e717dbc5caaf9600eb58bddc611834c63205dff5c2`
Expanded uniqueness receipt	`f014b8202ad435367824a5960387676bb3d568015e5e864f58cdde8c2bc069b1`
Expanded sorted index	`055fb9e007b0b7a261f4850b78450d1d8f306ec56e13c316cebf20514d1793ad`
Expanded-prefix manifest	`e26e618df5f47feabdf50534eccbe6fec7e1ba2b20581fedffec3f8a5b22a1b4`
False-unconditional control	`a8d615a2f5c8a2099d788820780c33531e44b05e8a5a81e8c357d57cf4227417`

The Markdown report in the same directory is the source-linked narrative. NAS paths and source identities are recorded there in full.